Checkpoint D — Architecture and Changeability

QuantRule

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1 Checkpoint D — Architecture and Changeability

1.1 Claim

The local-first architecture keeps the deterministic engine isolated from the desktop UI and the distribution channel, so a model, strategy, or UI change stays local and testable.

1.2 Current architecture

- One local quantum: Tauri UI + application service + pure engines.
- Pipes-and-filters inside a modular monolith.
- CLI and UI are adapters over the same application service.
- The download page is a static distribution channel.

1.3 Change scenarios

Change	Boundary	Evidence
New model reading	`quantrule-model`	golden, causality, determinism tests
New strategy rule	`quantrule-backtest`	strategy boundary tests
UI screen	`apps/desktop`	UI E2E
New installer	distribution	clean install
Team cloud (future)	new control plane	not implemented; new ADR

1.4 Evidence

- Crate boundaries and structural checks.

- Model/strategy boundary tests.
- The adapted Tauri UI will call the application service, not the engines.

1.5 Limit

The desktop and local research memory are not implemented yet. The team cloud is a future quantum and is not covered by this evidence.

1.6 Decision

Keep the pipeline/modular-monolith style; do not add a server or microservices for the solo increment.